

» Multiple Choice Questions (MCQs) «

DIRECTIONS : This section contains multiple choice questions. Each question has four choices (a), (b), (c) and (d) out of which only one is correct.

- The set which represents the isotope, isobar and isotone respectively is
 - $({}_1\text{H}^2, {}_1\text{H}^3)$, $({}_{79}\text{Au}^{197}, {}_{80}\text{Hg}^{198})$ and $({}_2\text{He}^3, {}_1\text{H}^2)$
 - $({}_2\text{He}^3, {}_1\text{H}^1)$, $({}_{79}\text{Au}^{197}, {}_{80}\text{Hg}^{198})$ and $({}_1\text{H}^1, {}_1\text{H}^3)$
 - $({}_2\text{He}^3, {}_1\text{H}^3)$, $({}_1\text{H}^2, {}_1\text{H}^3)$ and $({}_{79}\text{Au}^{197}, {}_{80}\text{Hg}^{198})$
 - $({}_1\text{H}^2, {}_1\text{H}^3)$, $({}_2\text{He}^3, {}_1\text{H}^3)$ and $({}_{79}\text{Au}^{197}, {}_{80}\text{Hg}^{198})$
- Which of the following statements are true/false for isotope?
 - Same number of protons but different number of neutrons.
 - Same number of neutrons but different number of protons.
 - Same number of protons and neutrons.
 - F, F, F
 - F, T, F
 - T, T, T
 - T, F, F
- If the radius of a nucleus of mass number 3 is R, then the radius of a nucleus of mass number 81 is
 - 3 R
 - 9 R
 - $(27)^{1/2}$ R
 - 27 R
- Which of the following statements are true/false?
 - Atoms of isotopes have same electronic structure.
 - Atoms of isotopes occupies same place in periodic table.
 - Atoms of isotopes have same number of protons.
 - Atoms of isotopes have same number of neutrons.
 - T, T, F, F
 - T, T, T, F
 - T, T, T, T
 - F, T, F, T
- Consider the following statements and select true/false.
 - The relative abundance of different isotopes differs from element to element.
 - Atomic species of the same element differing in mass but same number of protons are called isotopes.
 - Hydrogen has two isotopes.
 - T, F, F
 - F, T, F
 - T, T, F
 - T, T, T
- In a nuclear reactor, moderators slow down the neutrons which come out in a fission process. the moderator used have light nuclei. Heavy nuclei will not serve the purpose, because
 - they will break up
 - elastic collision of neutrons with heavy nuclei will not slow them down
 - the net weight of the reactor would be unbearably high
 - substances with heavy nuclei do not occur in liquid or gaseous state at room temperature
- If radius of the ${}_{13}^{27}\text{Al}$ nucleus is estimated to be 3.6 fermi then the radius of ${}_{52}^{125}\text{Te}$ nucleus be nearly
 - 8 fermi
 - 6 fermi
 - 5 fermi
 - 4 fermi
- The constituents of atomic nuclei are believed to be
 - neutrons and protons
 - protons only
 - electrons and protons
 - electrons, protons and neutrons.
- The mass number of a nucleus is
 - always less than its atomic number
 - always more than its atomic number
 - always equal to its atomic number
 - sometimes more than and sometimes equal to its atomic number

Nuclei

10. The volume of a nucleus is directly proportional to
(a) A (b) A^3 (c) $\sqrt{A}$ (d) $A^{1/3}$
11. Outside a nucleus
(a) neutron is stable
(b) proton and neutron both are stable
(c) neutron is unstable
(d) neither neutron nor proton is stable
12. A nucleus represented by the symbol ${}_Z^AX$ has
(a) A protons and $(Z-A)$ neutrons
(b) Z neutrons and $(A-Z)$ protons
(c) Z protons and $(A-Z)$ neutrons
(d) Z protons and A neutrons
13. Which of the following statement is incorrect?
(a) Atoms of isotopes have same electronic structure.
(b) Atoms of isotopes occupies same place in periodic table.
(c) Atoms of isotopes have same number of protons.
(d) Atoms of isotopes have same number of neutrons.
14. If the distance between nuclei is 2×10^{-13} cm, the density of nuclear material is
(a) 3.21×10^{-12} kg/m³ (b) 1.6×10^{-3} kg/m³
(c) 2×10^9 kg/m³ (d) 1×10^{17} kg/m³
15. Which of the following statement is incorrect ?
(a) Nuclear density is a constant for all matter.
(b) Nuclear density is around 3.3×10^{17} kg/m³.
(c) Nuclear density is very large compared to ordinary matter.
(d) Mass of ordinary matter is mainly due to nucleus.
16. If radius of the ${}_{12}^{27}\text{Al}$ nucleus is taken to be R_{Al} , then the radius of ${}_{53}^{125}\text{X}$ nucleus is nearly:
(a) $\frac{5}{3}R_{\text{Al}}$ (b) $\frac{3}{5}R_{\text{Al}}$
(c) $\left(\frac{13}{53}\right)^{1/3}R_{\text{Al}}$ (d) $\left(\frac{53}{13}\right)^{1/3}R_{\text{Al}}$
17. Which of the following statements is true for nuclear forces?
(a) they obey the inverse square law of distance
(b) they obey the inverse third power law of distance
(c) they are short range forces
(d) they are equal in strength to electromagnetic forces.
18. If in nuclear fusion process the masses of the fusing nuclei be m_1 and m_2 and the mass of the resultant nucleus be m_3 , then
(a) $m_3 > (m_1 + m_2)$ (b) $m_3 = m_1 + m_2$
(c) $m_3 = |m_1 - m_2|$ (d) $m_3 < (m_1 + m_2)$
19. Heavy water is used in nuclear reactors
(a) to absorb neutrons to sustain controlled reaction
(b) to absorb neutrons to stop the chain reaction
(c) to reduce hazardous radiation from nuclear reaction
(d) to slow down the neutrons to thermal energies
20. Which of the following nuclear reactions is not possible?
(a) ${}^{12}_6\text{C} + {}^{12}_6\text{C} \longrightarrow {}^{20}_{10}\text{Ne} + {}^4_2\text{He}$
(b) ${}^9_4\text{Be} + {}^1_1\text{H} \longrightarrow {}^6_3\text{Li} + {}^4_2\text{He}$
(c) ${}^{11}_5\text{Be} + {}^1_1\text{H} \longrightarrow {}^9_4\text{Be} + {}^4_2\text{He}$
(d) ${}^7_3\text{Li} + {}^4_2\text{He} \longrightarrow {}^1_1\text{H} + {}^{10}_4\text{B}$
21. In any fission process, the ratio $\frac{\text{mass of fission products}}{\text{mass of parent nucleus}}$ is
(a) equal to 1
(b) greater than 1
(c) less than 1
(d) depends on the mass of the parent nucleus
22. From the following equations, pick out the possible nuclear reactions.
(a) ${}_6^{13}\text{C} + {}^1_1\text{H} \rightarrow {}_6^{14}\text{C} + 4.3 \text{ MeV}$
(b) ${}_6^{12}\text{C} + {}^1_1\text{H} \rightarrow {}_7^{13}\text{N} + 2 \text{ MeV}$
(c) ${}_7^{14}\text{N} + {}^1_1\text{H} \rightarrow {}_8^{16}\text{O} + 7.3 \text{ MeV}$
(d) ${}_{92}^{235}\text{U} + {}^1_0\text{n} \rightarrow {}_{54}^{140}\text{X} + {}_{38}^{94}\text{Si} + {}^{20}_{10}\text{n} + \gamma + 200 \text{ MeV}$
23. A nuclei having same number of neutron but different number of protons / atomic number are called
(a) isobars (b) isomers (c) isotones (d) isotopes
24. The nuclei of which one of the following pairs of nuclei are isotones?
(a) ${}_{34}^{74}\text{Se}$, ${}_{31}^{71}\text{Ga}$ (b) ${}_{38}^{84}\text{Sr}$, ${}_{38}^{86}\text{Sr}$
(c) ${}_{42}^{92}\text{Mo}$, ${}_{40}^{92}\text{Zr}$ (d) ${}_{20}^{40}\text{Ca}$, ${}_{16}^{32}\text{S}$
25. For effective nuclear forces, the distance should be
(a) 10^{-10} m (b) 10^{-13} m
(c) 10^{-15} m (d) 10^{-20} m
26. The ratio of the mass densities of nuclei of ${}_{40}\text{Ca}$ and ${}_{16}\text{O}$ is
(a) 5 (b) 2 (c) 0.1 (d) 1
27. Nucleus of an atom whose atomic mass is 24 consists of
(a) 11 electrons, 11 protons and 13 neutrons
(b) 11 electrons, 13 protons and 11 neutrons
(c) 11 protons and 13 neutrons
(d) 11 protons and 13 electrons

28. A nucleus splits into two nuclear parts which have their velocity ratio equal to 2 : 1. What will be the ratio of their nuclear radius?
- (a) $2^{1/3} : 1$ (b) $1 : 2^{1/3}$
(c) $3^{1/2} : 1$ (d) $1 : 3^{1/2}$
29. Atomic weight of boron is 10.81 and it has two isotopes ${}_5\text{B}^{10}$ and ${}_5\text{B}^{11}$. Then ratio of ${}_5\text{B}^{10} : {}_5\text{B}^{11}$ in nature would be
- (a) 19 : 81 (b) 10 : 11
(c) 15 : 16 (d) 81 : 19
30. The nuclear radius of ${}_8\text{O}^{16}$ is 3×10^{-15} m. If an atomic mass unit is 1.67×10^{-27} kg, then the nuclear density is approximately
- (a) $2.35 \times 10^{17} \text{ g cm}^{-3}$ (b) $2.35 \times 10^{17} \text{ kg m}^{-3}$
(c) $2.35 \times 10^{17} \text{ gm}^{-3}$ (d) $2.35 \times 10^{17} \text{ kg mm}^{-3}$
31. Nuclear forces are
- (a) spin dependent and have no non-central part
(b) spin dependent and have a non-central part
(c) spin independent and have no non-central part
(d) spin independent and have a non-central part
32. Two nucleons are at a separation of 1 fermi. The net force between them is F_1 if both are neutrons, F_2 if both are protons and F_3 if one is proton and the other is a neutron. Then
- (a) $F_1 > F_2 > F_3$ (b) $F_1 = F_3 > F_2$
(c) $F_2 > F_1 > F_3$ (d) $F_1 = F_2 > F_3$
33. A moderator is used in nuclear reactors in order to
- (a) slow down the speed of the neutrons
(b) accelerate the neutrons
(c) increase the number of neutrons
(d) decrease the number of neutrons
34. Fusion reactions take place at high temperature because
- (a) atoms are ionised at high temperature
(b) molecules break up at high temperature
(c) nuclei break up at high temperature
(d) kinetic energy is high enough to overcome repulsion between nuclei
35. For a nuclear fusion process, suitable nuclei are
- (a) any nuclei
(b) heavy nuclei
(c) lighter nuclei
(d) nuclei lying in the middle of periodic table
36. Fusion reaction occurs at temperatures of the order of
- (a) 10^3 K (b) 10^7 K (c) 10 K (d) 10^4 K
37. Control rods used in nuclear reactors are made of
- (a) stainless steel (b) graphite
(c) cadmium (d) plutonium

38. Which of the following statements is true?

- (a) ${}_{78}\text{Pt}^{192}$ has 78 neutrons
(b) ${}_{84}\text{Po}^{214} \rightarrow {}_{82}\text{Pb}^{210} + \beta^-$
(c) ${}_{92}\text{U}^{238} \rightarrow {}_{90}\text{Th}^{234} + {}_2\text{He}^4$
(d) ${}_{90}\text{Th}^{234} \rightarrow {}_{91}\text{Pa}^{234} + {}_2\text{He}^4$

» Assertion & Reason

DIRECTIONS : Each of these questions contains an assertion followed by reason. Read them carefully and answer the question on the basis of following options. You have to select the one that best describes the two statements.

- (a) If both **Assertion** and **Reason** are **correct** and the Reason is the **correct explanation** of the Assertion.
(b) If both **Assertion** and **Reason** are correct but Reason is **not the correct explanation** of the Assertion.
(c) If the **Assertion** is **correct** but **Reason** is **incorrect**.
(d) If the **Assertion** is **incorrect** but the **Reason** is **correct**.

39. **Assertion :** Density of all the nuclei is same.

Reason : Radius of nucleus is directly proportional to the cube root of mass number.

40. **Assertion :** Nuclear forces are independent of charges.

Reason : Nuclear force is not a central force.

» Match the Following

DIRECTIONS : Each question contains statements given in two columns which have to be matched. Statements (A, B, C, D) in column-I have to be matched with statements (1, 2, 3, 4) in column-II.

41. Match the column-I and column-II.

Column – I	Column – II
(A) Nuclear fusion	(1) $E = mc^2$
(B) Nuclear fission	(2) Generally possible for nuclei with low atomic number
(C) β -decay	(3) Generally possible for nuclei with higher atomic number
(D) Mass-energy equivalence	(4) Essentially proceeds by weak reaction nuclear forces
(a) (A) $\rightarrow$ (2); (B) $\rightarrow$ (3); (C) $\rightarrow$ (4); (D) $\rightarrow$ (1)	
(b) (A) $\rightarrow$ (4); (B) $\rightarrow$ (1); (C) $\rightarrow$ (2); (D) $\rightarrow$ (4)	
(c) (A) $\rightarrow$ (1); (B) $\rightarrow$ (3); (C) $\rightarrow$ (2); (D) $\rightarrow$ (4)	
(d) (A) $\rightarrow$ (3); (B) $\rightarrow$ (4); (C) $\rightarrow$ (2); (D) $\rightarrow$ (1)	

» Fill in the Blanks

DIRECTIONS : Complete the following statements with an appropriate word / term to be filled in the blank space(s).

42. The nuclei ${}_6\text{C}^{13}$ and ${}_7\text{N}^{14}$ can be described as _____.
43. Order of magnitude of density of uranium nucleus is ($m_p = 1.67 \times 10^{-27} \text{ kg}$) _____ kg/m^3 .
44. The ratio of volumes of nuclei (assumed to be in spherical shape) with respective mass numbers 8 and 64 is _____.
45. If u denotes 1 atomic mass unit. One atom of an element has mass exactly equal to Au , where A is mass number of element and is equal to _____.
46. If the radius of a nucleus ${}^{256}\text{X}$ is 8 fermi, then the radius of ${}_2\text{He}$ nucleus will be _____ fermi.

» True / False

DIRECTIONS : Read the following statements and write your answer as true or false.

47. The nuclear density is independent of mass number.
48. Strong nuclear force is the strongest force in nature.
49. Nuclear force is charge dependent.
50. Radius of nucleus (R) is related to mass number (A) as $R = R_0 A^{1/2}$ where R_0 is a constant.

ANSWER KEY & SOLUTIONS

1. (d) ${}_1\text{H}^2$ and ${}_1\text{H}^3$ are isotopes
 ${}_2\text{He}^3$ and ${}_1\text{H}^3$ are isobars
 ${}_{79}\text{Au}^{197}$ and ${}_{80}\text{Hg}^{198}$ are isotones.
2. (d) For isotope Z is same and A is different. Therefore the number of neutrons A-Z will also be different.
3. (a)
4. (b) The nuclei having equal number of neutrons called isotones.
5. (c) Hydrogen has three isotopes.
6. (b) The moderator used have light nuclei (like proton). When protons undergo perfectly elastic collision with the neutron emitted their velocities are exchanged, i.e., neutrons come to rest and protons move with the velocity of neutrons. To slowdown the speed of neutrons substance should be made up of proton for perfectly elastic i.e., we need light nuclei not heavy nuclei because heavy nuclei will not serve the purpose because elastic collisions of neutrons with heavy nuclei will not slow them down or speed but only direction will change.
7. (b) $R = R_0(A)^{1/3}$

$$\therefore \frac{R_1}{R_2} = \left(\frac{A_1}{A_2}\right)^{1/3} = \left(\frac{27}{125}\right)^{1/3} = \frac{3}{5}$$

$$R_2 = \frac{5}{3} \times 3.6 = 6 \text{ fermi}$$
8. (a) Nucleus contains only neutrons and protons.
9. (d) In hydrogen, atomic number and mass number are equal.
10. (a) Radius of nucleus $R = R_0 A^{1/3}$ where A is the mass number of nucleus.

$$\therefore \text{Volume of nucleus} = \frac{4}{3} \pi R^3 = \left(\frac{4}{3} \pi R_0^3\right) A$$

$$\therefore \text{Volume is proportional to A.}$$
11. (c) Out side the nucleus, neutron is unstable (life ≈ 932 s).
12. (c) ${}_Z^AX$ has Z protons and (A - Z) neutrons
13. (d)
14. (d) Density of nuclear material = mass/volume

$$= \frac{10^{-27}}{\frac{4}{3} \pi r^3} = \frac{3 \times 10^{-27}}{4 \pi (2 \times 10^{-15})^3} = 10^{17} \text{ kg/m}^3$$
15. (b) As $R \propto A^{1/3}$ and volume of nucleus is proportional to R^3 is proportional to A. Thus, the density of nucleus is a constant, independent of A, for all nuclei. Different nuclei are like drop of liquid of constant density. The density of nuclear matter is approximately $2.3 \times 10^{17} \text{ kg m}^{-3}$.
 This density is very large compared to ordinary matter, say water which is 10^3 kg m^{-3} . This is understandable, as we have already seen that most of the atom is empty. Ordinary matter consisting of atoms has a large amount of empty space.
16. (a) As we know, $R = R_0(A)^{1/3}$
 where A = mass number
 $R_{\text{Al}} = R_0(27)^{1/3} = 3R_0$
 $R_X = R_0(125)^{1/3} = 5R_0 = \frac{5}{3} R_{\text{Al}}$
17. (c) Nuclear forces are short range attractive forces which balance the repulsive forces between the protons inside the nucleus.
18. (d) $m_3 < (m_1 + m_2)$ ($\because m_1 + m_2 = m_3 + E$)
 as $E = [m_1 + m_2 - m_3] C^2$
19. (d) Heavy water is used in nuclear reactors to slow down the neutrons to thermal energies to initiate the nuclear reaction.
20. (c) In this reaction mass is not conserved.
21. (c) Binding energy per nucleon for fission products is higher relative to Binding energy per nucleon for parent nucleus, i.e., more masses are lost and are obtained as kinetic energy of fission products. So, the given ratio < 1 .
22. (b) 23. (c)
24. (a) Isotones means equal number of neutrons i.e., (A-Z)
 $= 74 - 34 = 71 - 31 = 40$.
25. (c)
26. (d) Nuclear density is independent of atomic number.
27. (c) Nucleus does not contain electron.
28. (b) As momentum is conserved, therefore,

$$\frac{m_1}{m_2} = \frac{A_1}{A_2} = \frac{v_2}{v_1} = \frac{1}{2}$$

$$\therefore \frac{R_1}{R_2} = \left(\frac{A_1}{A_2}\right)^{1/3} = \left(\frac{1}{2}\right)^{1/3} = 1 : 2^{1/3}$$
29. (a) Let the percentage of B^{10} atoms be x, then average atomic weight

$$= \frac{10x + 11(100 - x)}{100} = 10.81 \Rightarrow x = 19$$

Nuclei

$$\therefore \frac{N_{B^{10}}}{N_{B^{11}}} = \frac{19}{81}$$

30. (b) For nucleus of ${}^8\text{O}^{16}$

$$\text{Mass} = (16)(1.67 \times 10^{-27}) \text{ kg}$$

$$\text{Volume} = \frac{4}{3}\pi R^3 = \frac{4}{3}\pi(3 \times 10^{-15})^3 \text{ m}^3 = 36\pi \times 10^{-45} \text{ m}^3$$

$$\text{Density} = \frac{\text{mass}}{\text{volume}} = \frac{16 \times 1.67 \times 10^{-27} \text{ kg}}{36\pi \times 10^{-45} \text{ m}^3} \\ = 2.35 \times 10^{17} \text{ kg m}^{-3}$$

31. (b)

32. (b) In case of Proton-Proton Electrostatic repulsive force is also present which reduces the net force.

33. (c) Moderator slows down neutrons.

34. (d) Extremely high temps needed for fusion make K.E. large enough to overcome repulsion between nuclei.

35. (c) 36. (b)

37. (c) Control rods are made of cadmium.

38. (c)

39. (a) Density, $\rho = \frac{M}{V} = \frac{A}{\frac{4}{3}\pi r^3}$

$$= \frac{A}{\frac{4}{3}\pi(r_0 A^{1/3})^3} = \frac{1}{\left(\frac{4}{3}\pi r_0^3\right)} = \text{constant}$$

40. (b)

41. (a) (A) $\rightarrow$ (2); (B) $\rightarrow$ (3); (C) $\rightarrow$ (4); (D) $\rightarrow$ (1)

42. (Isotones) As ${}^6\text{C}^{13}$ and ${}^7\text{N}^{14}$ have same no. of neutrons ($13 - 6 = 7$ for C and $14 - 7 = 7$ for N), so they are isotones.

43. (10^{17}) The order of magnitude of mass and volume of uranium nucleus will be

$$m \simeq A(1.67 \times 10^{-27} \text{ kg}) \quad (A \text{ is atomic number})$$

$$V = \frac{4}{3}\pi r^3 \simeq \frac{4}{3}\pi[(1.25 \times 10^{-15} \text{ m})A^{1/3}]^3 \\ \simeq (8.2 \times 10^{-45} \text{ m}^3)A$$

$$\text{Hence, } \rho = \frac{m}{V} = \frac{A(1.67 \times 10^{-27} \text{ kg})}{(8.2 \times 10^{-45} \text{ m}^3)A} \\ \simeq 2.0 \times 10^{17} \text{ kg m}^{-3}.$$

44. (0.125) As we know $R = R_0 A^{1/3}$

$$\therefore R \propto A^{1/3}$$

$$\text{As, } V \propto R^3 \text{ or } V \propto A$$

$$\therefore \frac{V_1}{V_2} = \frac{8}{64} = \frac{1}{8} = 0.125$$

45. (1) $A = 1$

46. (2) $R = R_0(A)^{1/3}$

$$\therefore \frac{R_1}{R_2} = \left(\frac{A_1}{A_2}\right)^{1/3} = \left(\frac{256}{4}\right)^{1/3} = 4$$

$$R_2 = \frac{R_1}{4} = \frac{8}{4} = 2 \text{ fermi}$$

47. (True) The order of nuclear density is 10^{17} .

48. (True)

49. (False)

50. (False) Radius $R = R_0 A^{1/3}$